ST. CHARLES PREPARATORY SCHOOL

# Honors Geometry — 3.5 Practice

Equations of Parallel and Perpendicular Lines • Sections 3.5 • Mr. Nappier

Name: _____ Period: _____ Date: _____

Work every problem in the space given — the work is the point, not the answer. After each part there is a green box that tells you whether you got them right, without giving the answers away. Use it before you move on.

## PART A — Slope first

*Everything in 3.5 starts with a slope. Get these right and the rest is bookkeeping.*

**Worked for you — follow these two steps every time.**

Find the slope through (1, −2) and (5, 6).

Step 1 — label the points: $x_1 = 1$, $y_1 = -2$, $x_2 = 5$, $y_2 = 6$

Step 2 — subtract in the SAME order, top and bottom: $m = (6 - (-2)) / (5 - 1) = 8/4 = 2$

**1.** Find the slope of the line through (−3, 8) and (5, −4).

m = (____ − ____) / (____ − ____) = _____

answer: _____

**2.** Find the slope of the line through (2, −7) and (2, 5).

m = (____ − ____) / (____ − ____) = _____

answer: _____

**3.** Find the slope of the line through (−6, 4) and (3, 4).

m = (____ − ____) / (____ − ____) = _____

answer: _____

**4.** A line with slope m = −2/5 passes through (−1, 6) and (9, ____). Find the missing coordinate.

−2/5 = (____ − 6) / (9 − (−1)) y = _____

answer: _____

**✓ Check yourself — Part A**

Your four answers should be, in some order: one negative fraction, one undefined, one zero, and one positive whole number. If you have two of anything, one of them is wrong — and #2 and #3 are the usual suspects.

## PART B — Parallel, perpendicular, coinciding, or neither

*Rewrite anything that is not already solved for y. Compare slopes, THEN compare y-intercepts.*

**The decision, in order.**

Same slope, different b → parallel. Same slope, same b → coinciding (one line wearing two outfits).

Slopes multiply to −1 → perpendicular. Anything else → neither.

A vertical line and a horizontal line are always perpendicular.

**1.** $y = \frac{3}{4}x - 2$ and $y = \frac{3}{4}x + 5$

$m_1$ = _____ $m_2$ = _____ → _____

answer: _____

**2.** y = −5x + 1 and x − 5y = 15

rewrite: y = _____ → _____

answer: _____

**3.** 2x + 3y = 12 and y = −⅔x + 4

rewrite: y = _____ → _____

answer: _____

**4.** y = 2x + 7 and y = −2x + 7

$m_1$ = _____ $m_2$ = _____ → _____

answer: _____

**5.** x = −4 and y = 9

describe each line: _____ → _____

answer: _____

**6.** 6x − 2y = 8 and y = 3x + 1

rewrite: y = _____ → _____

answer: _____

**✓ Check yourself — Part B**

When you are done you should have exactly two parallel, two perpendicular, one coinciding, and one neither. If your tally is off, the one to re-read is whichever problem you did NOT rewrite into y = mx + b first.

## PART C — Write the equation

*Three moves, every time: get the slope of the given line, adjust it (same for parallel, negative reciprocal for perpendicular), then push it through the point.*

**Worked for you — write the line through (3, 1) parallel to y = 2x − 9.**

Step 1 — slope of the given line: m = 2. Step 2 — parallel, so the new slope is also 2.

Step 3 — point-slope, then solve for y: y − 1 = 2(x − 3) → y − 1 = 2x − 6 → y = 2x − 5

Step 4 — check: does (3, 1) satisfy it? 2(3) − 5 = 1 ✓

**1.** Through (4, −3), parallel to y = ½x + 6.

answer: _____

**2.** Through (−6, 1), perpendicular to y = 3x − 8.

answer: _____

**3.** Through (2, 9), parallel to 4x + y = 5.

answer: _____

**4.** Through (−3, −2), perpendicular to 2x − 5y = 10.

answer: _____

**5.** Through (0, −7), parallel to y = −x + 4.

answer: _____

**6.** Through (5, 1), perpendicular to y = ⅛x − 3.

answer: _____

**Answer Bank**

| | | |
|---|---|---|
| y = −4x + 17 **(O)** | y = ½x + 5 **(T)** | y = ½x − 5 **(S)** |
| y = −x − 7 **(E)** | y = −3x − 1 **(A)** | y = −5/4 x + 29/4 **(S)** |
| y = −⅓x − 1 **(L)** | y = −4x + 9 **(R)** | y = −5/2 x − 19/2 **(P)** |

**✓ Check yourself — Part C**

Find each of your six answers in the Answer Bank and write its letter in the boxes below, in order 1 through 6. They spell one word — a word this whole section is about. Three bank entries are decoys; if you land on one, that problem is wrong and you know exactly which.

| 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|
| | | | | | |

## PART D — Push further

*Same three moves, hiding behind a sentence.*

**1.** Error analysis. Jo says the line through (1, 4) perpendicular to y = 2x + 3 is y = 2x + 2. Jo is wrong. Name the mistake in one sentence, then write the correct equation.

answer: _____

**2.** Write the equation of the line through (−2, 7) that is parallel to the x-axis. Then write the equation of the line through (−2, 7) that is perpendicular to the x-axis.

answer: _____

**3.** Line a passes through (−1, 2) and (3, k). Line a is perpendicular to y = −2x + 1. Find k.

answer: _____

**4.** Line m is y = ⅓x + 2. Line n is perpendicular to m and passes through (6, −1). Do m and n cross on the y-axis? Find where they cross and justify your answer.

answer: _____

**✓ Check yourself — Part D**

Add your k from #3 to the x-coordinate of the crossing point in #4. You should get 8.5. If you do not, exactly one of those two is wrong — and #3 is the faster one to redo.

*Finished early? Go back to any problem where you wrote the answer but not the work, and write the work. That is the half of the test that earns partial credit.*